# The Silent Golden Station

## The Harmonic Fingerprint of the Cube–Octahedron Sweep Finds Icosaxes, and Only Icosaxes

Theorem: along the sweep from the octahedron to the cube, the cubic harmonic of degree 4 vanishes at one solid only, Icosaxes at μ = φ, where every degree with no icosahedral harmonic falls silent; degree 8 vanishes there and at one bronze station

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Paper #94 · Version 2 · 7 November 2026 (Version 1: 4 November 2026) · DICTO DOI: 10.5281/zenodo.23934349 (concept) · Licence CC BY-SA 4.0 · Scripts, checks and figure: https://github.com/DICTOR-Master/kaleidohedra/tree/research/studies/2026-11-04-harmonic-fingerprint, 2026-11-04-co-rd-sweep, 2026-11-04-mahler-sweep, 2026-11-04-tetra-sweep and 2026-11-04-harmonic-4d (research branch)
The test came from DICTO's widening of the search to harmonics, waves and resonance (The DICTO Concordance, Part 5) and his call to run it. DICTO proposed the parallel outside tests of §5, unrelated measures run on the same sweep so that each checks the others, and the test of whole four-dimensional structures in §7. Computation, proofs and drafting with Claude (Anthropic).

**Abstract.** Give every corner of a solid seen from its centre a unit weight on the sphere and expand in spherical harmonics: the strength $Q_l$ of each degree l is the solid's harmonic fingerprint, the bond-orientational order of Steinhardt, Nelson and Ronchetti [1], which physicists use to tell the cubic, hexagonal and icosahedral order of atoms apart. We apply it to #90's sweep $P_\mu = C \cap \mu C^*$ of the cube and its polar octahedron, from the octahedron (μ = 1) to the cube (μ = 3). **Main theorem.** (A) $Q_4$ vanishes at exactly one point of the sweep, μ = φ, Icosaxes; there the corner condition is $a^4 - 3a^2 + 1 = (a^2 + a - 1)(a^2 - a - 1) = 0$ with $a = \mu - 1$. (B) At Icosaxes $Q_l$ vanishes exactly for l = 2, 4, 8, 14, the even degrees with no icosahedral harmonic: its 24 corners are the two icosahedra of the pyritohedral pair (#90), so the one cubic solid on the sweep sounds icosahedral. (C) $Q_8$ vanishes at exactly two points, μ = φ and $\mu = (\sqrt{13} - 1)/2$, the reciprocals of the golden and the bronze ratio. (D) On the second dual sweep, the cuboctahedron cut by the rhombic dodecahedron, $Q_4$ vanishes once, at a point with no golden number, $1 + \sqrt{((6\sqrt{30} - 31)/17)}$. Version 2 adds two outside tests on the same sweep and two more walks: Mahler's volume product is 32/3 at both ends and largest at a cubic irrational $\mu^* \approx 1.6312$, just past Icosaxes but not golden; the roundest member is $\mu = \sqrt{3}$, and in n dimensions $\mu = \sqrt{n}$, by Lindelöf's theorem on polytopes circumscribed about a ball; on the Delian walk the cubic tone falls silent once, at t = 1/9, never with degree 8; on #91's four-dimensional sweep $Q_4$ vanishes only at the 24-cell and $Q_6$ exactly at $\mu = \sqrt{2}$ and $1 + \sqrt{3}$. The proofs are short polynomial identities; every value is checked numerically against the classical fingerprints of the cubic, hexagonal and body-centred packings and the icosahedron.

*A solid can be listened to: its corners, seen from the centre, are notes on a sphere, and each degree of spherical harmonic is a tone. Cubic solids ring in the fourth tone, icosahedral ones never do. Sliding from the octahedron to the cube, the fourth tone falls silent once, at*

> *exactly the golden station, Icosaxes, because its corners are two icosahedra. The sound finds the golden point with no geometry put in.*

## 1. The fingerprint

For a finite set of directions $u_1, \dots, u_N$ put $q_{lm} = (1/N)\,\Sigma\, Y_{lm}(u_i)$ and $Q_l = (4\pi/(2l+1)\,\Sigma_m\,|q_{lm}|^2)^{1/2}$ [1]. $Q_l$ does not change under rotation; $\hat{W}_l$ is the normalised third-order invariant built with Wigner 3j symbols [12]. Since Frank's observation that liquids favour icosahedral order [6] and the polytetrahedral picture of Nelson and Spaepen [7], these numbers have become the standard local test of order in simulations of crystals, liquids and glasses [8, 9], with known limits [10]. For a set with the full cube–octahedron symmetry every $q_{lm}$ lies in the space of cube-invariant harmonics of degree l, which for l = 4, 6, 8 is one-dimensional, spanned (Meyer [11]) by $K_4 = x^4 + y^4 + z^4 - (3/5)r^4$, $K_6$ and $K_8 = \Sigma x^8 - 14\Sigma x^6y^2 + 35\Sigma x^4y^4$ (sums over the distinct terms; harmonic, checked). Hence **$Q_l = 0$ exactly when $\Sigma\, K_l(u_i) = 0$**, which turns each zero into a polynomial equation.

| Corner set | $Q_4$ | $Q_6$ | $Q_8$ | $Q_{10}$ | $Q_{12}$ |
|---|---|---|---|---|---|
| Cuboctahedron (cubic close packing) | 0.191 | 0.575 | 0.404 | 0.013 | 0.600 |
| Anticuboctahedron (hexagonal close packing) | 0.097 | 0.485 | 0.317 | 0.010 | 0.565 |
| Body-centred shell (8 + 6) | 0.036 | 0.511 | 0.429 | 0.195 | 0.405 |
| Icosahedron | 0 | 0.663 | 0 | 0.363 | 0.585 |
| Dodecahedron | 0 | 0.368 | 0 | 0.560 | 0.020 |
| Cube (= tetrahedron) | 0.509 | 0.628 | 0.213 | 0.650 | 0.415 |
| Octahedron | 0.764 | 0.354 | 0.718 | 0.411 | 0.696 |

The first four rows are the classical values [1] and serve as the check of the code. The table shows two limits of the method: the even tones do not hear a point reflection, so the tetrahedron sounds exactly like the cube (its corners and their mirror images are the cube's), and the rhombic dodecahedron's corner directions are the body-centred shell's. The third-order invariant $\hat{W}_6$ does separate the dual pair dodecahedron (+0.170) and icosahedron (−0.170): the same tone, opposite sign, corners against faces.

## 2. Main theorem

> **Main theorem.** Let $P_\mu = C \cap \mu C^*$, $C = [-1, 1]^3$, $C^* = \{|x| + |y| + |z| \le 1\}$, $1 < \mu < 3$ (#90 [2]).
> (A) $Q_4(P_\mu) = 0$ if and only if $\mu = \varphi$ (Icosaxes).
> (B) At $\mu = \varphi$, $Q_l = 0$ for l = 2, 4, 8, 14 and $Q_l \neq 0$ for the other even $l \le 16$.
> (C) $Q_8(P_\mu) = 0$ if and only if $\mu = \varphi$ or $\mu = (\sqrt{13} - 1)/2$.
> (D) On the sweep $CO \cap \mu RD$ from the rhombic dodecahedron ($\mu = 1$) to the cuboctahedron ($\mu = 2$), with $CO = \{|x| \le 1, |x| + |y| + |z| \le 2\}$ and RD its polar, $Q_4 = 0$ at exactly one point, $\mu_0 = 1 + a_0$, $17a_0^4 + 62a_0^2 - 7 = 0$, so $a_0^2 = (6\sqrt{30} - 31)/17$ and $\mu_0 \approx 1.3311$.
> (E) On the tetrahedron's own dual sweep $T \cap \mu(-T)$ (truncated tetrahedra, octahedron at $\mu = 1$), $Q_4 = 0$ exactly at the root $\mu \approx 0.5538$ of $9\mu^4 - 52\mu^3 + 30\mu^2 - 4\mu + 1 = 0$ and at its mirror $1/\mu$; neither is the golden cut ($\mu = (3 + 2\sqrt{5})/11$ or $(\varphi + 2)/(3\varphi + 2)$) nor the regular truncated tetrahedron ($\mu = 3/5$).

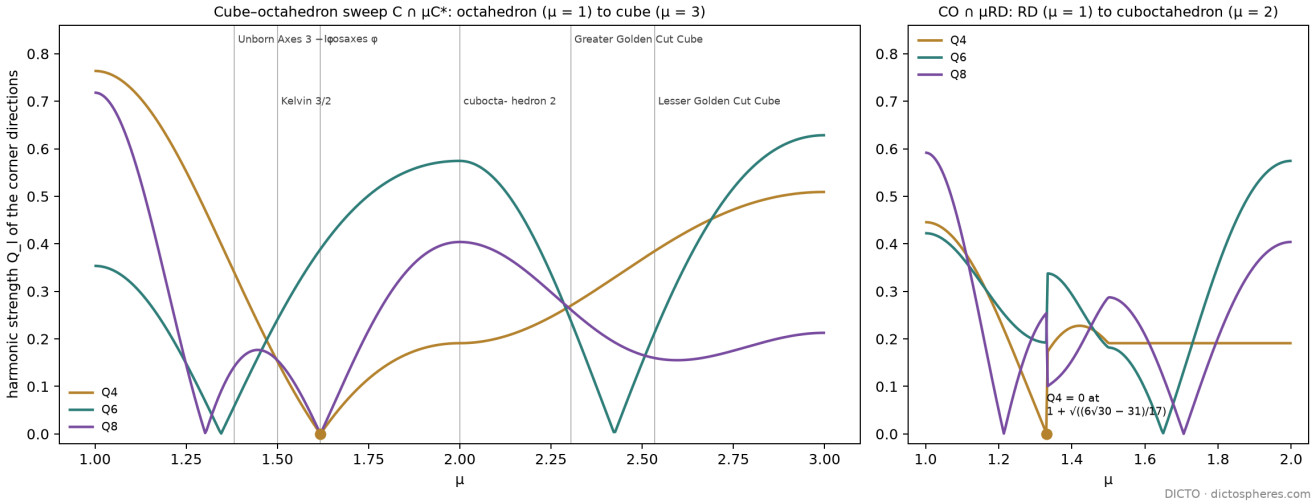



**Figure 1.** The fingerprint along the two sweeps. Left: from the octahedron to the cube, with Unborn Axes, Kelvin's cell, Icosaxes, the cuboctahedron and the two Golden Cut Cubes marked; $Q_4$ (gold) and $Q_8$ (violet) touch zero together only at Icosaxes, and $Q_8$ once more at the bronze station $\mu = (\sqrt{13} - 1)/2 \approx 1.303$. Right: from the rhombic dodecahedron to the cuboctahedron; $Q_4$ touches zero once, at a point with no golden number.

*Proof.* For $1 < \mu \le 2$ the corners of $P_\mu$ are the 24 points $(\pm 1, \pm a, 0)$ in all orders, $a = \mu - 1$; for $2 \le \mu < 3$ the 24 points $(\pm 1, \pm 1, \pm b)$, $b = \mu - 2$ (#90). (A) For (1, a, 0), $K_4/r^4 = (1 + a^4)/(1 + a^2)^2 - 3/5$, which vanishes when $2a^4 - 6a^2 + 2 = 0$, that is $a^4 - 3a^2 + 1 = (a^2 + a - 1)(a^2 - a - 1) = 0$; in (0, 1) only $a = 1/\varphi$, $\mu = \varphi$. For (1, 1, b), $(2 + b^4)/(2 + b^2)^2 = 3/5$ gives $b^4 - 6b^2 - 1 = 0$, whose positive root $b^2 = 3 + \sqrt{10}$ is above 1. (B) At $a = 1/\varphi$ the corners are the cyclic arrangements of $(0, \pm 1, \pm 1/\varphi)$ in both orders, the vertices of two icosahedra (the compound of #90, Figure 1 (c) [2]). Each icosahedron alone has $q_{lm} = 0$ in every degree that carries no icosahedral harmonic, and the even degrees that carry one are 0, 6, 10, 12, 16, 18, … (the degrees $6p + 10q$ [3]); so the mean over both vanishes at $l = 2, 4, 8, 14$. The non-vanishing at $l = 6, 10, 12, 16$ is the computed value (0.389, 0.313, 0.357, 0.403). (C) With $K_8$, the sum over the corners (1, a, 0) is $6(a^2 + a - 1)(a^2 - a - 1)(a^2 + 3a - 1)(a^2 - 3a - 1)$; in (0, 1) the roots are $a = 1/\varphi$ and $a = (\sqrt{13} - 3)/2 = 1/\beta_3$, $\beta_3 = (3 + \sqrt{13})/2$ the bronze ratio. On (2, 3) the sum over the corners (1, 1, b) is $3(b^8 - 28b^6 + 70b^4 - 28b^2 + 9)$, which has no root in (0, 1); $Q_8$ stays above 0.155. (D) For $1 < \mu < 4/3$ the corners are the 24 points $(\pm 1, \pm a, \pm a)$ in all orders and the 8 points $(\pm \mu/2, \pm \mu/2, \pm \mu/2)$ (computed face census: 6 squares and 12 hexagons). The eight give $K_4/r^4 = -4/15$ each, so $Q_4 = 0$ when $24((1 + 2a^4)/(1 + 2a^2)^2 - 3/5) = 32/15$, i.e. $17a^4 + 62a^2 - 7 = 0$. On (4/3, 2) no zero was found on a grid of 2000 points; from $\mu = 3/2$ on, $Q_4$ stays at the cuboctahedron's 0.1909 to four decimals. ∎

## 3. What the sound hears

> *The cubic tone and the eighth tone fall silent together only where the corners become two icosahedra, and the eighth tone alone hears one more of the metallic ratios, the bronze.*

The fourth-degree condition $a^4 - 3a^2 + 1 = 0$ is the golden equation twice, for a and for $-a$; nothing else on the sweep can silence the cubic tone. The eighth-degree condition adds the bronze pair $a^2 \pm 3a - 1$, the next metallic ratio after the golden one with the same form $x^2 = nx + 1$ ($n = 1$ golden, $n = 3$ bronze); the silver ratio ($n = 2$), which carries the Golden Cut Cubes' second field in #90, does not appear. The bronze station $\mu = (\sqrt{13} - 1)/2$ is not one of the solids named so far; its corners are $(1, 1/\beta_3, 0)$ and it sits between the octahedron and

Unborn Axes. $Q_6$ also vanishes twice on the sweep, at $a^2 = (17 - \sqrt{273})/4$ ($\mu \approx 1.345$) and at $\mu \approx 2.424$, near Unborn Axes and the truncated cube but at neither.

The second dual sweep, from the rhombic dodecahedron to the cuboctahedron, has no such station: its one zero of $Q_4$ involves $\sqrt{30}$ and no golden number. So the icosahedral sound belongs to the first pair, the cube and the octahedron, where #48 and #90 found Icosaxes and Unborn Axes in the first place.

## 4. Networks: every node by its own sound

The fingerprint was made for networks: physicists compute it at every atom from its nearest neighbours to tell the local order. The same works on our networks. On the stacking ABCBACB, repeated, the twelve neighbours of each ball give $Q_4 = 0.191$ at every straight ball and 0.097 at every twisted one, so the fingerprint reads off the c/h word of the Hexagonality Law papers ball by ball, with no knowledge of the layers (checked on all seven letters). In a network of pieces it does the same for any node: the set of directions to its neighbours, or to the corners of the pieces around it, has a fingerprint that does not depend on how the node is turned, so equal nodes are found wherever they sit, and icosahedral nodes ($Q_4 = Q_8 = 0$) stand out among cubic ones.

## 5. Three outside tests on one sweep

> *A sound is one way to judge a solid. Two others come from far away: how much room a solid and its polar leave between them (convex geometry), and how round it is for its surface (soap films). DICTO proposed to run such unrelated tests side by side on the same sweep, so that each checks the others. The sound finds Icosaxes; roundness finds the solid whose faces all touch one ball, for a reason known since 1869; the volume product finds a point just past Icosaxes that has no golden number in it.*

For a convex body K symmetric about the origin, Mahler's volume product is $f(K) = |K| \cdot |K^\circ|$, $K^\circ$ the polar body; it does not change under linear maps, and in three dimensions its least value 32/3 is taken by the cube and the octahedron (Mahler [19]; proved by Iriyeh and Shibata [20]). The isoperimetric quotient $IQ = 36\pi V^2/A^3$ is 1 for the ball and less for every other solid; in n dimensions the same role is played by $V^{n-1}/A^n$.

> **Proposition 1 (the volume product along the sweep).** Let $f(\mu) = |P_\mu| \cdot |P_\mu^\circ|$ for $P_\mu = C \cap \mu C^*$, $1 \le \mu \le 3$.
> (a) $f(\mu) = 32(\mu^3 - 3(\mu - 1)^3)/(3\mu^2) = -32(2\mu^3 - 9\mu^2 + 9\mu - 3)/(3\mu^2)$ for $1 \le \mu \le 2$, and $f(\mu) = (4/\mu)(8 - (4/3)(3 - \mu)^3)$ for $2 \le \mu \le 3$.
> (b) $f = 32/3$ at both ends, the octahedron and the cube; $f = 128/9$ at Kelvin's cell ($\mu = 3/2$), 40/3 at the cuboctahedron ($\mu = 2$) and $(544 - 224\sqrt{5})/3 \approx 14.3736$ at Icosaxes ($\mu = \varphi$).
> (c) f is largest at the root $\mu^*$ of $2\mu^3 - 9\mu + 6 = 0$ in (1, 2), $\mu^* = \sqrt{6}\cos(\tfrac{1}{3}\arccos(-\sqrt{(2/3)})) \approx 1.6312$, with $f(\mu^*) \approx 14.3751$. The point $\mu^*$ is a cubic irrational, not $\varphi$, and f is symmetric neither under the core turn $\mu \leftrightarrow 3/\mu$ nor under the pair swap $\mu \leftrightarrow 3 - \mu$.

*Proof.* For $1 \le \mu \le 2$, $P_\mu$ is the octahedron $\mu C^*$ (volume $4\mu^3/3$) with six corner pyramids of height $\mu - 1$ cut off by the cube's faces, each of volume $2(\mu - 1)^3/3$; its polar $\mathrm{conv}(C^*, C/\mu)$ is the cube $C/\mu$ with a square pyramid of height $1 - 1/\mu$ on each face, of volume $8/\mu^3 + 6 \cdot (1/3)(2/\mu)^2(1 - 1/\mu) = 8/\mu^2$. For $2 \le \mu \le 3$, $P_\mu$ is the cube with eight corner tetrahedra with legs $3 -$

μ cut off, volume $8 - (4/3)(3 - \mu)^3$, and its polar is the octahedron with a tetrahedral cap of volume $(3/\mu - 1)/6$ on each of its eight faces, $4/3 + (4/3)(3/\mu - 1) = 4/\mu$. This gives (a), and (b) by substitution. On (1, 2), $f'(\mu) = -32(2\mu^3 - 9\mu + 6)/(3\mu^3)$, which changes sign once, from + to −, at μ*; on (2, 3), $f'(\mu) = 16(2\mu^3 - 9\mu^2 + 21)/(3\mu^2)$, with one root $\mu \approx 2.0853$ in the interval, a lower local maximum $f \approx 13.388$. The trigonometric form is Viète's solution of the depressed cubic $\mu^3 - (9/2)\mu + 3 = 0$. ∎

The ends are exactly Mahler's minimum, as they must be: the test passes its calibration. Its maximum lies within 1% of Icosaxes but is not golden, so the volume product does not hear the golden station that the sound hears.

> **Proposition 2 (the roundest member, by Lindelöf's theorem).** For $n \geq 2$ let $S_\mu = [-1, 1]^n \cap \{|x_1| + \ldots + |x_n| \leq \mu\}$, $1 < \mu < n$, the sweep from the cross-polytope to the cube. Then $V^{n-1}/A^n$ is largest at $\mu = \sqrt{n}$ and at no other point. For $n = 2$ the roundest member is the regular octagon; for $n = 3$ it is $P_{\sqrt{3}}$, with IQ = 0.78164; for $n = 4$ it is the 24-cell.

*Proof.* Lindelöf's theorem [16], extended to every dimension by Minkowski [17] (see also [18]): among the convex polytopes whose facet normals all belong to a given finite set N of directions, the ratio $V^{n-1}/A^n$ is largest for the polytope circumscribed about a ball with one facet normal to each direction of N, and only for it and its homothetic copies. Every member of the sweep has its facet normals among the $2n$ coordinate directions and the $2^n$ directions $(\pm 1, \ldots, \pm 1)/\sqrt{n}$, at distances 1 from the centre for the cube's facets and $\mu/\sqrt{n}$ for the cross-polytope's. At $\mu = \sqrt{n}$ both distances are 1: $S_{\sqrt{n}}$ is circumscribed about the unit ball, and since $1 < \sqrt{n} < n$ every one of the $2n + 2^n$ directions carries a facet. So $S_{\sqrt{n}}$ is Lindelöf's optimum for this set of normals. No other member is homothetic to it, because its two distances differ. For $n = 4$, $S_2$ has the 24 corners $(\pm 1, \pm 1, 0, 0)$ in all orders, the 24-cell. ∎

The studies confirm the theorem from the other side: in three dimensions, for $1 < \mu < 2$, $V = 4\mu^3/3 - 4(\mu - 1)^3$ and $A = 12(\mu - 1)^2 + 2\sqrt{3}(2\mu^2 - 6(\mu - 1)^2)$, and $d \log IQ/d\mu$ is $-18\mu(\mu - 1)(\mu - \sqrt{3})$ times a positive factor; for $n = 2, \ldots, 12$ the derivative vanishes exactly at $\sqrt{n}$, and a numerical scan finds no higher value elsewhere on the sweep. This also explains why the core turn of #81 [23] has the fixed point $\sqrt{3}$: the turn swaps the two facet distances (#90 [2]), so its fixed point is where they are equal, Lindelöf's point. In the same way the octahedron, where the faces of T and −T are equally far from the centre, is the roundest solid on the tetrahedron's sweep $T \cap \mu(-T)$ of Main theorem (E).

| Test (field) | Special point on $P_\mu$ | Named station? |
|---|---|---|
| $Q_4 = 0$ (bond order, physics) | $\mu = \varphi$ | Icosaxes, exact (Main theorem (A)) |
| $Q_8 = 0$ | $\varphi$ and $(\sqrt{13} - 1)/2$ | Icosaxes and the bronze station (C) |
| $Q_6 = 0$ | $\approx 1.345$ and $\approx 2.424$ | none (§3) |
| Roundness maximum (soap films) | $\mu = \sqrt{3}$ | Lindelöf's point, the core turn's fixed point (Proposition 2) |
| Volume product minimum (convex geometry) | $\mu = 1$ and 3 | octahedron and cube, 32/3 (Proposition 1) |
| Volume product maximum | $\mu^* \approx 1.6312$ | none; within 1% of Icosaxes (Proposition 1) |

Each test lands on a named station only where a known reason puts it there: the sound at Icosaxes, because its corners are two icosahedra; roundness at √3, because of Lindelöf's theorem; the volume product at the ends, because of Mahler's minimum. Where no such reason exists, the extreme falls at an unnamed point. The second dual sweep shows the same.

> **Proposition 3 (the second sweep has no roundest station).** On the sweep CO ∩ μRD, 1 ≤ μ ≤ 2, of Main theorem (D):
> (a) the volume product is 40/3 at both ends, the rhombic dodecahedron and the cuboctahedron (a polar pair), and largest, f ≈ 15.387, at the root μ ≈ 1.5997 of $12\mu^3 - 36\mu^2 + 43 = 0$;
> (b) the isoperimetric quotient is largest at μ ≈ 1.5227, with IQ ≈ 0.8525;
> (c) no member with 1 < μ < 2 is circumscribed about a sphere, so Lindelöf's theorem names no member of this sweep.

*Computation and proof.* (a) The volume V and the polar volume V° are cubic in μ on three pieces, identified from computed hulls by exact fitting (to $10^{-9}$) and checked to agree at the joins: on (1, 4/3), $V = -6\mu^3 + 24\mu^2 - 24\mu + 8$ and $V^\circ = 4(3\mu + 2)/(3\mu^3)$; on (4/3, 3/2), $V = -24\mu^3 + 96\mu^2 - 120\mu + 152/3$ and $V^\circ = 6/\mu^2$; on (3/2, 2), $V = 8\mu^3 - 48\mu^2 + 96\mu - 172/3$ and $V^\circ = 4/\mu$. The ends give 2 · 20/3 = 40/3. The derivative of the product is $-8(3\mu^4 - 4\mu^2 - 8\mu + 8)/\mu^4$ on the first piece (no root there), $-16(9\mu^3 - 45\mu + 38)/\mu^3$ on the second (a lower local maximum ≈ 15.246 at μ ≈ 1.4331) and $16(12\mu^3 - 36\mu^2 + 43)/(3\mu^2)$ on the third. (b) is numerical. (c) The cube's facets $|x_i| = 1$ are present on the whole sweep, so a ball touching all facets has radius 1 and centre 0. For 1 < μ < 4/3 the other facets are the rhombic dodecahedron's, at distance μ/√2 < 1; for μ > 4/3 the corners (μ/2, μ/2, μ/2) are cut by the cuboctahedron's triangles |x| + |y| + |z| = 2, at distance 2/√3 ≠ 1. ∎

So on the second sweep neither roundness nor the volume product lands on a station, and Lindelöf's theorem says why for roundness: the cuboctahedron's own two kinds of face are at different distances, and no member makes all its faces touch one sphere. The exact point √3 of the first sweep is particular to the cube–octahedron pair, for the classical reason, not by coincidence.

## 6. The Delian walk

> *The second walk through cubic solids, the Delian walk of #51, has a silent point too, but it is a plain fraction, and the eighth tone never joins it: no icosahedral order is hidden on this walk.*

Paper #51 [21] cuts every double cone of the octet by one plane at fraction t and gives the two sides to their centres; for 0 < t < 1/3 the octahedral piece O(t) is Kepler's rhombic dodecahedron with its eight three-fold corners cut by the planes s · x = 3(1 − t)/2, s = (±1, ±1, ±1), and the tetrahedral piece T(t) is a regular tetrahedron; for 1/3 < t < 1, O(t) is a regular octahedron and T(t) the cube with four alternate corners cut (the definitions of #93 [25]).

> **Proposition 4 (the Delian walk is cubic-silent once).** On 0 < t < 1/3, $Q_4(O(t)) = 0$ exactly at t = 1/9. $Q_8(O(t))$ vanishes near t ≈ 0.095 and t ≈ 0.205 and nowhere else on the

> study's grid, so never together with $Q_4$. The tetrahedral piece T(t) is never silent in degree 4; its $Q_6$ vanishes near $t \approx 0.560$ and $t \approx 0.774$.

*Proof.* The 30 corners of O(t) are the 6 points $(\pm 1, 0, 0)$ in all orders and the 24 points $(\pm p, \pm q, \pm q)$ in all orders, $p = (1 + 3t)/2$, $q = (1 - 3t)/2$. The six give $K_4/r^4 = 2/5$ each; the 24 give $(p^4 + 2q^4)/(p^2 + 2q^2)^2 - 3/5$ each. The sum is $-4(t - 1)(3t + 1)^2(9t - 1)/(9t^2 - 2t + 1)^2$, whose only zero in (0, 1/3) is $t = 1/9$, where the 24 corners point along (2, 1, 1). The zeros of $Q_8$ and the values for T(t) are computed on the study's grid. ∎

On the cube–octahedron sweep $Q_4$ and $Q_8$ vanish together at Icosaxes because two icosahedra hide there. On the Delian walk they never vanish together, so the walk has no hidden icosahedral station; its one cubic silence sits at a rational point.

## 7. Four dimensions

> *The same listening works on the 3-sphere, for the corners of four-dimensional solids. On #91's sweep from the 16-cell to the tesseract the cubic tone falls silent only at the 24-cell, the same solid that Lindelöf's theorem makes the roundest.*

For corner directions $u_i$ on the 3-sphere, $Q_l^2$ is the mean over all pairs of $U_l(u_i \cdot u_j)/(l + 1)$, $U_l$ the Chebyshev polynomial of the second kind (the zonal harmonic of degree l on the 3-sphere). For a corner set with the full symmetry of the tesseract, the invariant harmonics of degree 4 and of degree 6 each span one dimension, spanned by $K_4 = \Sigma x_i^4 - r^4/2$ and $K_6 = \Sigma x_i^6 - (5/4)r^2\Sigma x_i^4 + (5/16)r^6$ (harmonic, checked), so again $Q_l = 0$ exactly when $\Sigma\, K_l(u_i) = 0$. The calibration of this four-dimensional test on the regular polytopes (the 600-cell and the 120-cell are silent in every degree from 1 to 11, spherical 11-designs as is classical [24]) belongs to the Alicia in Wonderland volume and is reported there.

> **Proposition 5 (the four-dimensional sweep).** Let $S_\mu = K \cap \mu K^*$, $K = [-1, 1]^4$, $K^*$ the 16-cell $\{|x_1| + \ldots + |x_4| \le 1\}$, $1 < \mu < 4$ (#91 [22]).
> (a) $Q_4(S_\mu) = 0$ if and only if $\mu = 2$, the 24-cell, the fixed point of the core turn $\mu \leftrightarrow 4/\mu$.
> (b) $Q_6(S_\mu) = 0$ if and only if $\mu = \sqrt{2}$ or $\mu = 1 + \sqrt{3}$.
> (c) Neither vanishes at the All-Golden stations $\mu = \varphi^2$ and $\mu = 2 + 1/\varphi^2$ of #91.

*Proof.* The corners are the points $(\pm 1, \pm a, 0, 0)$ with $a = \mu - 1$ on (1, 2), $(\pm 1, \pm 1, \pm b, 0)$ with $b = \mu - 2$ on (2, 3) and $(\pm 1, \pm 1, \pm 1, \pm c)$ with $c = \mu - 3$ on (3, 4), each in all orders. (a) Per corner, $K_4/r^4$ is $(a^2 - 1)^2/(2(1 + a^2)^2)$, $b^2(b^2 - 4)/(2(2 + b^2)^2)$ and $(c^4 - 6c^2 - 3)/(2(3 + c^2)^2)$ on the three pieces; the only zero in the open pieces and their joins is $a = 1$, $b = 0$, which is $\mu = 2$, where the corners are $(\pm 1, \pm 1, 0, 0)$ in all orders. (b) Summed over the orders, $K_6/r^6$ is proportional to $(a^2 - 2a - 1)(a^2 + 2a - 1)/(1 + a^2)^2$ on (1, 2), to $(b^2 - 2)(b^2 - 2b - 2)(b^2 + 2b - 2)/(2 + b^2)^3$ on (2, 3) and to $(c^6 - 15c^4 + 75c^2 + 3)/(3 + c^2)^3$ on (3, 4), with positive constants. In (0, 1) the only roots are $a = \sqrt{2} - 1$ ($\mu = \sqrt{2}$) and $b = \sqrt{3} - 1$ ($\mu = 1 + \sqrt{3}$); the last factor is positive on [0, 1]. (c) follows from (a) and (b), since $\varphi^2 \approx 2.618$ and $2 + 1/\varphi^2 \approx 2.382$ are none of these points. ∎

At $\mu = \sqrt{2}$ the corner coordinate is $a = \sqrt{2} - 1 = 1/(1 + \sqrt{2})$, the reciprocal of the silver ratio, which the degree-4 and degree-8 conditions of the three-dimensional sweep do not contain (§3). The All-Golden stations are not heard: their golden content lives in the shapes of their

cells, not in the directions of their corners, and that is outside what this test can see. In four dimensions the sound and roundness agree: the 24-cell, silent through degree 5, is also the roundest member of the sweep (Proposition 2, n = 4).

## 8. Prior art, and the status of each result

**Known, credited.** The invariants $Q_l$, $\hat{W}_l$ and their classical values for the cubic, hexagonal and body-centred shells and the icosahedron: Steinhardt, Nelson and Ronchetti [1]. The degrees of icosahedral harmonics: Klein's invariants of degrees 6, 10, 15, and [3]. The sweep and its stations: #90 [2], #48 [4]; Icosaxes as the hull of two icosahedra: #90 [2]. The metallic ratios are classical [13]. The idea of hearing the solids is Kepler's [14]; the icosahedral cluster of Mackay [15] is the physical icosahedral shell. Version 2: Mahler's volume product and its minimum 32/3 at the cube and the octahedron, Mahler [19] and Iriyeh and Shibata [20]; the roundness of polytopes circumscribed about a ball, Lindelöf [16] and Minkowski [17] (see [18]), so Proposition 2 is a classical theorem read on our sweep, not a new result; the spherical designs of the regular polytopes, Delsarte, Goethals and Seidel [24]; the Delian walk, #51 [21] and #93 [25]; the four-dimensional sweep and its All-Golden stations, #91 [22]; the core turn, #81 [23].

**Proved here.** (A)–(E). We did not find in the sources checked a fingerprint of the cube–octahedron truncation family, nor the golden and bronze zeros. The classical uses fingerprint the neighbour shells of atoms; here the same numbers are read along a one-parameter family of solids, where they pick out its golden station. Version 2: Propositions 1, 3, 4 and 5 (the volume product along both sweeps, the Delian walk's silent point t = 1/9, the four-dimensional zeros), and the exact checks of Proposition 2 for n = 2, …, 12 on our sweeps. The sources checked found none of these computed for these families.

**Sources checked for Version 2.** The reference library (kaleidohedra-references, INDEX) was searched for Lindelöf, Minkowski, Fejes Tóth, isoperimetric, Mahler, Iriyeh, Santaló, Steinhardt, Bannai, Goethals and Delsarte. It holds Minkowski 1897 [17] as a link only (not read), and on isoperimetry only papers on a different question, the isoperimetric problem among parallelohedra (Lángi 2021, arXiv 2102.11621; Cesaroni and Novaga 2026, arXiv 2603.27221; Novaga and Song 2026, arXiv 2610.01653); none of these mentions Lindelöf. It has no entry for Lindelöf, Mahler or Iriyeh–Shibata: [16], [19] and [20] are cited from their standard bibliographic records, and [20] as found for The DICTO Concordance, Part 5.5. The study notes used: studies 2026-11-04-mahler-sweep, 2026-11-04-tetra-sweep §3, 2026-11-04-harmonic-4d and 2026-11-04-harmonic-fingerprint § Delian walk.

**Open.** The fingerprint of the golden nests and of the OG shells against the icosahedral clusters of quasicrystals; the odd degrees (l = 3, 7, 9), which hear the tetrahedron's mirror image; the reason $Q_4$ is constant from μ = 3/2 to 2 on the second sweep; the sweep from the icosidodecahedron to the rhombic triacontahedron; the exact roundness maximum and the exact zeros of $Q_8$ on the second sweep and the Delian walk; the volume product along the four-dimensional sweep.

## Data and reproducibility

studies/2026-11-04-harmonic-fingerprint: fingerprint.py (all $Q_l$, $\hat{W}_6$, the sweep, data/fingerprint.log, .json), fig_sweeps.py (Figure 1); studies/2026-11-04-co-rd-sweep:

co_rd_sweep.py (face census and $Q_4$ on the second sweep). Version 2: studies/2026-11-04-mahler-sweep (README: the exact volumes and products of Propositions 1 and 3, the roundness formula); studies/2026-11-04-tetra-sweep (sqrt_n.py and data/sqrt_n.json, the exact check n = 2, …, 12 of Proposition 2; sweep4d.py, data/sweep4d.json; measures.py, the tetrahedron sweep); studies/2026-11-04-harmonic-4d (README: Proposition 5 and the calibration); studies/2026-11-04-harmonic-fingerprint, README § Delian walk (Proposition 4). Run on dicto-big-brother with Python 3, numpy, scipy and sympy (exact polynomials for $K_4$, $K_6$, $K_8$).

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